

2.4 One-Sided Limits

In this section we extend the limit concept to one-sided limits, which are limits as x approaches the number c from the left-hand side (where $x < c$) or the right-hand side ($x > c$) only.

Recall

$\lim_{x \rightarrow c} f(x) = L \Leftrightarrow$ (a) as x approaches c from both sides of c ($x \neq c$), $f(x)$ gets closer and closer to L .

$$(b) \forall \varepsilon > 0, \exists \delta > 0, \exists 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

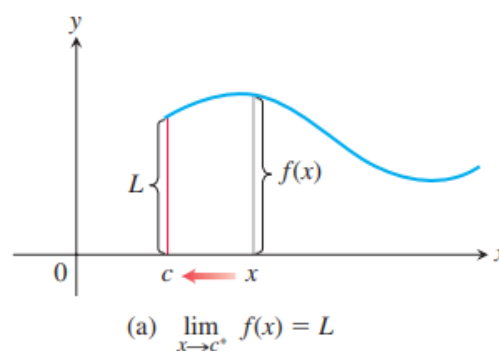
① Right-hand limit

$$\lim_{x \rightarrow c^+} f(x) = L \Leftrightarrow$$

(i) As $x \rightarrow c$ with $x > c$, $f(x)$ gets closer and closer to a single number L .

(ii) $\forall \varepsilon > 0, \exists \delta > 0$, such that

$$0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$



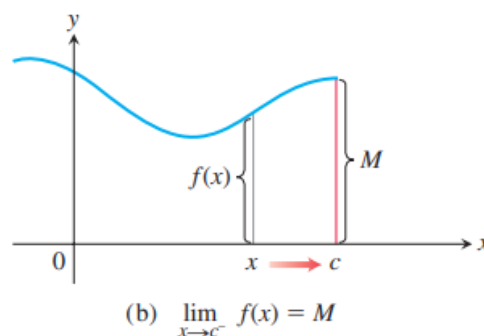
② Left-hand limit

$$\lim_{x \rightarrow c^-} f(x) = M \Leftrightarrow$$

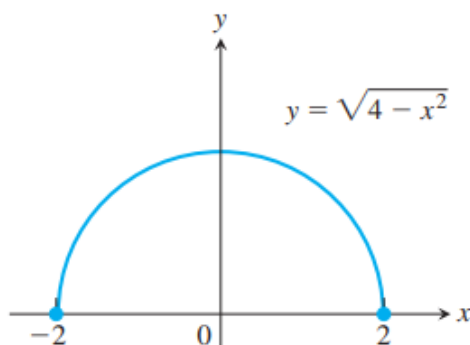
(i) As $x \rightarrow c$ with $x < c$, $f(x)$ gets closer and closer to a single number M .

(ii) $\forall \varepsilon > 0, \exists \delta > 0$, such that

$$-\delta < x - c < 0 \Rightarrow |f(x) - M| < \varepsilon.$$



Example 1 The domain of $f(x) = \sqrt{4 - x^2}$ is $[-2, 2]$; its graph is the semicircle in Figure.



We have

$$\lim_{x \rightarrow -2^-} f(x) = 0 \text{ and } \lim_{x \rightarrow -2^+} f(x) = 0.$$

The function does not have a left-hand limit at $x = -2$ or a right-hand limit at $x = 2$. It does not have a two-sided limit at either -2 or 2 because each point does not belong to an open interval over which f is defined.

Note 1 When the domain of f is an interval lying to the left of c , such as $(a, c]$ or (a, c) , then we say that f has a limit at c if it has a left-hand limit at c .

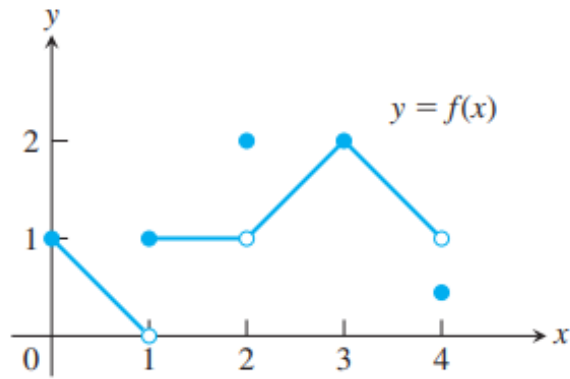
Similarly, if the domain of f is an interval lying to the right of c , such as $[c, b)$ or (c, b) , then we say that f has a limit at c if it has a right-hand limit at c .

Example 2 Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$.

Note 2 One-sided limits have all the properties listed in Theorem 1 in Section 2.2.

Note 3 $\lim_{x \rightarrow c} f(x) = L \Leftrightarrow \lim_{x \rightarrow c^+} f(x) = L \text{ and } \lim_{x \rightarrow c^-} f(x) = L$.

Example 3 Find the one-side limit and limit for the following function at points $x = 0, x = 1, x = 2, x = 3$ and $x = 4$.



Example 4

a) Graph function

$$f(x) = \begin{cases} 1-x^2, & x \neq 1 \\ 2, & x = 1 \end{cases}.$$

b) Find $\lim_{x \rightarrow 1^+} f(x)$ and $\lim_{x \rightarrow 1^-} f(x)$.

c) Does $\lim_{x \rightarrow 1} f(x)$ exist? If so, what is it? If not, why not?

Example 5 Find $\lim_{x \rightarrow -2^-} \left(\frac{x}{x+1} \right) \left(\frac{2x+5}{x^2+x} \right)$.

Example 6 Find $\lim_{x \rightarrow 1^+} \frac{|x-1|}{x-1}$.

Example 7 Find $\lim_{x \rightarrow 1^-} \frac{|x-1|}{x-1}$.

Example 8 Find $\lim_{x \rightarrow 0^+} \frac{\sqrt{x+4}-2}{x}$.

Note 4 $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$

Proof Let $0 < \theta < \frac{\pi}{2}$. Note that

Area $\triangle OAP < \text{area sector } OAP < \text{area } \triangle OAT.$

Since

$$\text{Area } \triangle OAP = \frac{1}{2} \sin \theta,$$

$$\text{area sector } OAP = \frac{\theta}{2},$$

$$\text{area } \triangle OAT = \frac{1}{2} \tan \theta = \frac{1}{2} \frac{\sin \theta}{\cos \theta},$$

we have

$$\frac{1}{2} \sin \theta < \frac{1}{2} \theta < \frac{1}{2} \frac{\sin \theta}{\cos \theta}.$$

That is,

$$\cos \theta < \frac{\sin \theta}{\theta} < 1.$$

Since $\lim_{\theta \rightarrow 0^+} \cos \theta = 1$, by the Sandwich Theorem, we have

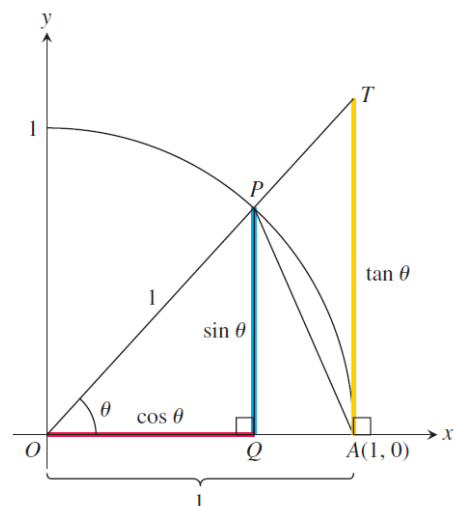
$$\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1.$$

But

$$\lim_{\theta \rightarrow 0^-} \frac{\sin \theta}{\theta} \stackrel{\alpha = -\theta}{=} \lim_{\alpha \rightarrow 0^+} \frac{\sin(-\alpha)}{-\alpha} = \lim_{\alpha \rightarrow 0^+} \frac{\sin \alpha}{\alpha} = 1.$$

Hence,

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$



Example 9 Show that (a) $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$ and (b) $\lim_{x \rightarrow 0} \frac{\sin 2x}{5x} = \frac{2}{5}.$

Example 10 Find $\lim_{t \rightarrow 0} \frac{\tan t \sec 2t}{3t}.$